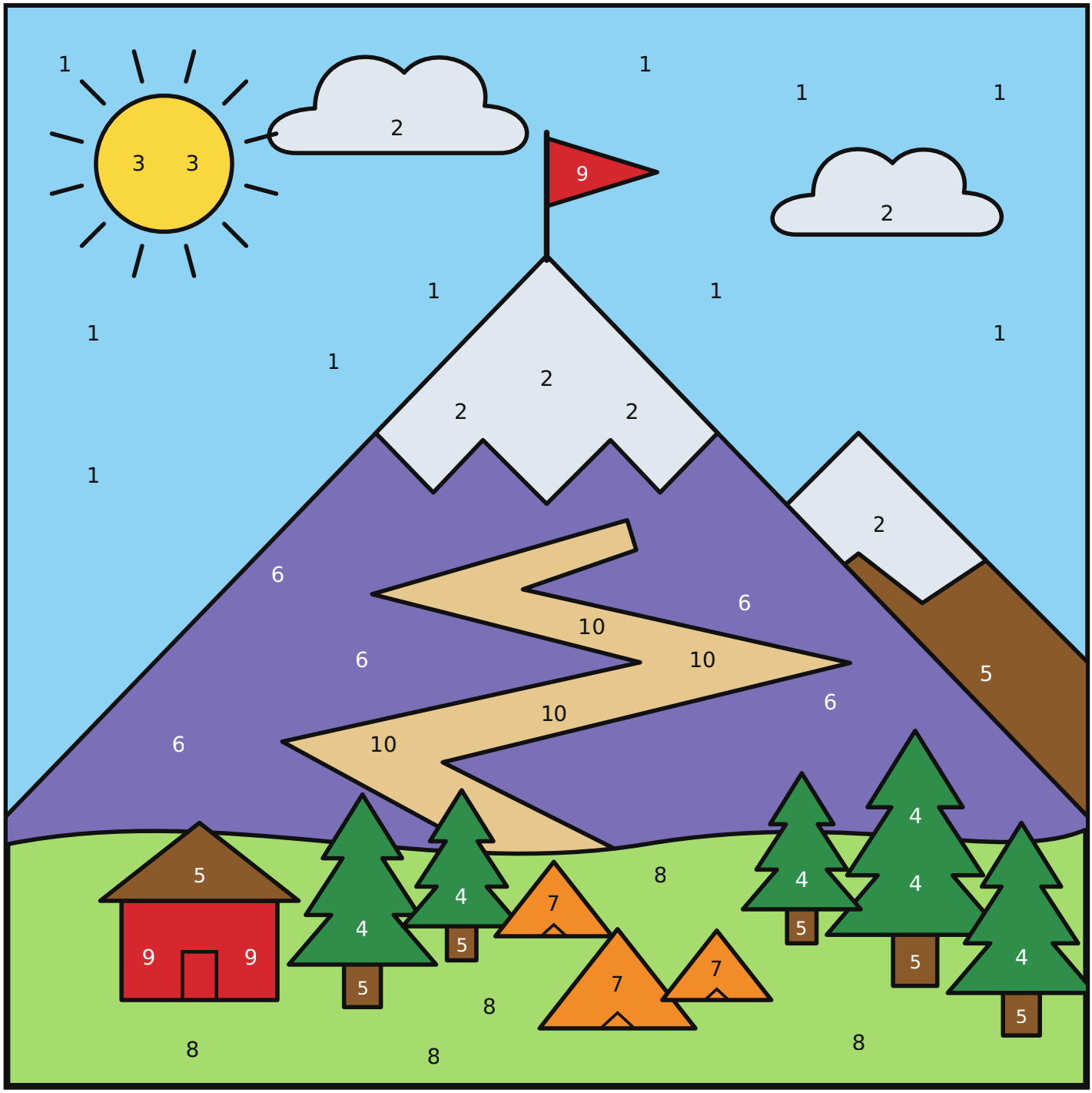
Name: **ANSWER KEY** _____ Date: _____

Right Triangles & Trigonometry: Color By Number



Color Code:

9 Red 30°	**6** Purple $\frac{3}{5}$	**8** Light Green $5\sqrt{3}$ ft	**7** Orange $\frac{3}{4}$	**5** Brown $4\sqrt{3}$
1 Light Blue 10	**3** Yellow $5\sqrt{2}$	**4** Green $4\sqrt{2}$	**2** Gray 12	**10** Tan 7.0

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Right Triangles & Trigonometry: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
A right triangle has legs 6 and 8. Find the hypotenuse.

$6^2 + 8^2 = 36 + 64 = 100$
10

1. Color: **Light Blue**

2.
A right triangle has a leg of 5 and a hypotenuse of 13. Find the other leg.

$5^2 + b^2 = 13^2$
$b^2 = 144$
12

2. Color: **Gray**

3.
An isosceles right triangle has legs of length 5. Find the hypotenuse.

45°-45°-90°: hypotenuse = leg · √2
5√2

3. Color: **Yellow**

4.
In a 45°-45°-90° triangle the hypotenuse is 8. Find the length of a leg.

$\text{leg} = \frac{\text{hypotenuse}}{\sqrt{2}} = \frac{8}{\sqrt{2}}$
rationalize: $\frac{8\sqrt{2}}{2}$
4√2

4. Color: **Green**

5.
In a 30°-60°-90° triangle the shortest side is 4. Find the longer leg.

longer leg = short leg · √3
4√3

5. Color: **Brown**

6.
In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find sin A.

$\sin A = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB}$
$\frac{3}{5}$

6. Color: **Purple**

7.
In right triangle ABC with the right angle at C, AB = 5, BC = 3 and AC = 4. Find tan A.

$\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{BC}{AC}$
$\frac{3}{4}$

7. Color: **Orange**

8.
A 10-ft ladder leans against a wall and makes a 60° angle with the ground. How high up the wall does it reach?

$\sin 60° = \frac{h}{10}$
$h = 10 \cdot \frac{\sqrt{3}}{2}$
5√3 ft

8. Color: **Light Green**

9.
If $\sin \theta = \frac{1}{2}$ for an acute angle θ, find θ.

$\theta = \sin^{-1}(\frac{1}{2})$
30°

9. Color: **Red**

10.
Find x to the nearest tenth: $\tan 35° = \frac{x}{10}$

$x = 10 \cdot \tan 35°$
7.0

10. Color: **Tan**